

CONTACT

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Tiruppur, Tamilnadu - 641667

Dinesh B

EDUCATION

2019 - 2023

PSG COLLEGE OF TECHNOLOGY, COIMBATORE

Bachelor of Engineering (B.E.) in Electronics and Communication Engineering

CGPA - 9.46/10

2017 - 2019

VEVEAHAM MATRIC HR. SEC. SCHOOL, DHARAPURAM

96.8%

2015 - 2017

KANNAMMAL NATIONAL MATRIC HR. SEC. SCHOOL, PALLADAM

98.6%

SKILLS

- Languages: C, C++, Embedded C, Python
- Embedded Tools: Code Composer Studio, T32 Debugger, CANoe
- Standards: AUTOSAR, ISO 26262
- Protocols: UDS, CAN, SPI, DoIP, Ethernet
- Version Control: Git, Bitbucket, Gerrit
- Build Systems: CMake,
- Testing Tools: CANTATA, CppUnit, T32, EA, Doors

LANGUAGES

- English (Fluent)
- Tamil (Fluent)
- Telugu (Basics)
- Hindi (Intermediate)

DINESH B

EMBEDDED SOFTWARE ENGINEER

I am a proactive and self-motivated engineer with a strong passion for creating impactful solutions in the technology space. Leading multiple initiatives during my academic journey has sharpened my communication, organizational, and team leadership abilities. I am recognized for my reliability and my ability to present innovative ideas that drive meaningful progress. Always eager to learn and collaborate, I strive to contribute effectively to every project I undertake.

WORK EXPERIENCE

Bosch Global Software Technologies, Coimbatore

Volkswagen (VW) ADAS — OCT 2023 - JAN 2024

Bootloader & Software Update (OTA)

Worked on Bootloader and OTA flashing features for ECUs using UDS diagnostics over Ethernet. Developed and integrated components for Fast Bootloader and Software Updater, within AUTOSAR architecture, focusing on diagnostics and system robustness.

AUDI GEN2 - Keyless Anchor — FEB 2024 - APR 2024

Software Update - Keyless Entry

Developed firmware for UWB Ranging & Localization on the TI CC2340R5 MCU, enabling Keyless Entry functionality with precise positioning. Implemented SPI-based BLE communication, CAN-based inter-module interaction, and a robust Software Update pipeline using Code Composer Studio and UniFlash.

MPCI - Functional Safety — APR 2024 - SEP 2024

Functional Safety

Developed and tested Temperature Monitoring Unit (TMU) functionality, including fault injection, and safe state handling. Designed and executed unit tests using the CANTATA framework and Rhapsody for safety-critical modules such as TMU, eMCeM, BIST, PMIC, and PMC.

Driver Monitoring System (DMS) — OCT 2024 - MAR 2025

Functional Safety

Implemented the safety module SM119 in Stellantis and updated compilers across ECUs to ensure compatibility and enhance performance. Developed and integrated unit tests for SSH components, which were later extended to Daimler Truck and IVECO projects. Utilized tools such as CppUnit, CANoe, T32 Debugger, Gerrit to maintain traceability and ensure accurate debugging.

Fusion Demonstrator — APR 2025 - PRESENT

Cluster Applications

Developed and validated Instrument Cluster modules such as Speedometer, Tachometer, Telltales, and WAI (Warnings and Indicators). Achieved seamless integration with platform stacks including ClusterApp, CMake, SCP, and DocAsCode for consistent delivery and maintainability.

INTERNSHIPS

Samsung Research Institution **(6 Months - 2021)**

Worked on AI/ML-based alarm correlation systems, improving automated fault detection and reducing noise in alert management.

Bosch Global Software Technologies **(2 Months - 2022)**

Performed Inverter Testing using HIL modeling, and ECU Test Tool; automated tests using Python.

Bosch Global Software Technologies **(4 Months - 2023)**

Gained hands-on experience in ECU testing and test automation, ensuring robust validation of automotive embedded systems. Learned CAN and FlexRay protocols for effective communication across vehicle ECUs.